

## NONPARAMETRIC TESTS FOR TREATMENT EFFECTS AFTER A PRELIMINARY TEST ON BLOCK EFFECTS IN A RANDOMIZED BLOCK DESIGN

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*Summary:* For a randomized block design, the problem of testing for treatment effects following a preliminary test for block effects is considered. A nonparametric procedure is formulated and the effect of the preliminary test on the size and power of the ultimate test is studied. The proposed nonparametric procedure is also compared with its parametric counterpart and the Pitman-efficiency of the procedure is calculated. The small sample properties of the test are examined by means of a Monte Carlo study.

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### 1. INTRODUCTION

A variety of statistical techniques which consist of a preliminary test being applied prior to the actual inference being made have been developed, e.g. preliminary test estimators and testing hypotheses after application of a preliminary test. Among the earliest contributions are those of Bancroft (1944) and Bechhofer (1951). A variety of such procedures in

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ANOVA are discussed in Bancroft and Han (1980), and Saleh and Sen (1982) proposed a nonparametric procedure in a regression model.

In this paper we consider a two-way ANOVA based on the model

$$X_{ijh} = \mu + \alpha_i + \beta_j + \epsilon_{ijh}, \quad i=1, \dots, m; \quad j=1, \dots, k; \quad h=1, \dots, c$$

where  $\mu$ ,  $\alpha_i$  and  $\beta_j$  are constant, row and column effects respectively,  $\sum_i \alpha_i = 0$ ,  $\sum_j \beta_j = 0$ , and  $\epsilon_{ijh}$  is a random variable from an unknown distribution with continuous distribution function (d.f.)  $F$  defined on the real line. We consider the hypothesis  $H_{10}: \beta_1 = \dots = \beta_k$  that the treatments are the same, but we do not know whether there are block effects, i.e. whether  $H_{20}: \alpha_1 = \dots = \alpha_m$  holds.

Clearly if  $H_{20}$  holds, the model reduces to  $X_{ijh} = \mu + \beta_j + \epsilon_{ijh}$  and  $H_{10}$  can be tested by means of the Kruskal-Wallis test, whereas if  $H_{20}$  does not hold, the appropriate test is that of Friedman. If the Friedman test is used in the absence of block effects, information is lost and it will therefore not be as powerful as the Kruskal-Wallis test. In many cases the loss of power can be quite large and it would be desirable to use the best test available. We therefore propose the following preliminary test (pretest) procedure: First test  $H_{20}$  at level  $\alpha_2^*$ . If  $H_{20}$  is rejected, use the Friedman test for  $H_{10}$  and if  $H_{20}$  is not rejected use the Kruskal-Wallis test. Thus it is attempted to use the best test under the prevailing circumstances.

Clearly the size of differences between block effects and by implication  $\alpha_2^*$  could play a role in the pretest procedure. The procedure is examined for observations drawn from a variety of distributions with treatment and block effects of different sizes.

The results show that the pretest procedure is in many cases just as or asymptotically just as good as the "best" test and significantly better than the alternative test. Compared to the corresponding parametric pretest procedure it is just as or slightly more robust and more powerful for most non-normal distributions.

Thus the nonparametric pretest procedure is not only theoretically a good procedure to use, but has practical advantages over other procedures.

## 2. THE PROPOSED TEST PROCEDURE

Denote the d.f. of  $X_{ijh}$  by  $F_{ij}$ , i.e.  $F_{ij}(x) = F(x - \mu - \alpha_i - \beta_j)$ . To obtain a contiguous alternative hypothesis we define

$$F_{ij,N}(x) = F(x - \mu - \alpha_i - \frac{\beta_j}{\sqrt{N}}) = F_{j,N}(x - \alpha_i) = F_i(x - \frac{\beta_j}{\sqrt{N}}),$$

where  $N = mkc$  and  $N \rightarrow \infty$  if and only if  $c \rightarrow \infty$ .

We test  $H_{20}$  against  $H_{2a,N}$ :  $F_{ij,N}(x) = F_{j,N}(x - \alpha_i)$  (i.e.  $\alpha_i \neq \alpha_{i'}$  for at least one pair  $i \neq i'$ ) by means of the Friedman test, based on the test statistic

$$T_2 = \frac{12}{mkc^2(mc+1)} \sum_i \left[ \sum_j \sum_h (R_{ijh} - \frac{mc+1}{2}) \right]^2$$

with ranks allocated per column. Under  $H_{20}$ ,  $T_2$  has a  $\chi^2(m-1)$  distribution. The test based on  $T_2$  is consistent against  $H_{2a,N}$ .

If  $H_{20}$  is rejected, we test  $H_{10}$  against  $H_{1a,N}$ :  $F_{ij,N}(x) = F_i(x - \frac{\beta_j}{\sqrt{N}})$  by means of

$$T_1 = \frac{12}{mkc^2(kc+1)} \sum_j \left[ \sum_i \sum_h (R_{ijh} - \frac{kc+1}{2}) \right]^2$$

with ranks now allocated per row. From Mack and Skillings (1980) it follows that  $T_1$  has asymptotically under  $H_{1a,N}$ , for  $N \rightarrow \infty$  (i.e.  $c \rightarrow \infty$ ), a  $\chi^2(k-1, \lambda_1^2)$  distribution, i.e. a non-central  $\chi^2$  distribution with  $(k-1)$  degrees of freedom and non-centrality parameter  $\lambda_1^2$ . Under the conditions of Lemma 7.2 of Puri (1964), the non-centrality parameter is given by

$$\lambda_1^2 = \frac{12}{k} \left( \int_{-\infty}^{+\infty} f^2(x) dx \right)^2 \sum_{j=1}^k \beta_j^2.$$

If  $H_{20}$  is not rejected, we test  $H_{10}$  against  $H_{1a,N}^*$ :  $F_{ij,N}(x) = F(x - \mu - \frac{\beta_j}{\sqrt{N}})$

by means of the Kruskal-Wallis test with test statistic

$$K = \frac{12}{N(N+1)} \sum_j R_{.j}^2 / mc - 3(N+1),$$

where ranks are now allocated to all values jointly and  $R_{.j} = \sum_i \sum_h R_{ijh}$ . Note that if these ranks have been allocated, it is easy to obtain ranks per column (for  $T_2$ ) and per row (for  $T_1$ ). Under  $H_{1a,N}^*$  and conditions given in, for instance, Puri and Sen (1971) p. 135,  $K$  has asymptotically a  $\chi^2(k-1, \lambda_1^2)$  distribution.

If  $c_2$  denotes the upper  $\alpha_2^u$  critical value of  $T_2$ , the overall test statistic can be expressed as

$$TK = T_1 \cdot I(T_2 \geq c_2) + K \cdot I(T_2 < c_2) \text{ with } I(\cdot) \text{ the indicator function.}$$

The parametric procedure is as follows: Test  $H_{20}$  against  $H_{2a,N}$  using

$$F_2 = \frac{kc \sum_i (\bar{X}_{i..} - \bar{X}_{...})^2 / (m-1)}{\sum_i \sum_j \sum_h (X_{ijh} - \bar{X}_{i..} - \bar{X}_{.j.} + \bar{X}_{...})^2 / (mkc - m - k + 1)}$$

where  $\bar{X}_{i..} = \frac{1}{kc} \sum_j \sum_h X_{ijh}$ , etc. Under  $H_{20}$   $F_2$  has an  $F(m-1, mkc - m - k + 1)$  distribution and the test based on  $F_2$  is consistent against  $H_{2a,N}$ .

If  $H_{20}$  is rejected, test  $H_{10}$  against  $H_{1a,N}$  using

$$F_1 = \frac{mc \sum_j (\bar{X}_{.j.} - \bar{X}_{...})^2 / (k-1)}{\sum_i \sum_j \sum_h (X_{ijh} - \bar{X}_{i..} - \bar{X}_{.j.} + \bar{X}_{...})^2 / (mkc - m - k + 1)}$$

which has an  $F(k-1, mkc - m - k + 1)$  distribution under  $H_{10}$ . Under  $H_{1a,N}$  and for  $\sigma^2 = \text{var}(X_{ijh})$  finite,  $(k-1)F_1$  has asymptotically, for  $N \rightarrow \infty$ , a  $\chi^2(k-1, \lambda_1^2)$  distribution with

$$\lambda_1^2 = \frac{1}{k\sigma^2} \sum_j \beta_j^2$$

(cf. Mack and Skillings (1980)).

If  $H_{20}$  is not rejected, test  $H_{10}$  against  $H_{1a,N}^*$  using

$$U = \frac{mc \sum_j (\bar{X}_{.j} - \bar{X}_{...})^2 / (k-1)}{\sum_{ijh} (X_{ijh} - \bar{X}_{.j})^2 / k(mc-1)}$$

which has, under  $H_{10}$ , an  $F(k-1, k(mc-1))$  distribution and under  $H_{1a,N}^*$   $(k-1)U$  has asymptotically, for  $N \rightarrow \infty$ , a  $\chi^2(k-1, \lambda_1'^2)$  distribution.

The final parametric test statistic is therefore

$$FU = F_1 \cdot I(F_2 \geq e_2) + U \cdot I(F_2 < e_2)$$

with  $e_2$  the  $\alpha_2''$  upper critical point of the  $F(m-1, mkc-m-k+1)$  distribution.

### 3. THE SIZE AND POWER OF TK

Let  $\underline{\alpha} = (\alpha_1, \dots, \alpha_m)$  denote the vector of row effects and  $\underline{\beta} = (\beta_1, \dots, \beta_k)$  the vector of column effects. If we denote the chosen level of significance of  $T_2(T_1, K)$  by  $\alpha_2''(\alpha_1'', \alpha_3'')$ , its upper critical value by  $c_2(c_1, c_3)$  and its size by  $\alpha_2'(\alpha_1', \alpha_3')$  when using the proposed pretest procedure, the size of TK is

$$\begin{aligned} \alpha_N'(\underline{\alpha}; \underline{\beta} = \underline{0}) &= P(\text{reject } H_{10} | \underline{\alpha}; \underline{\beta} = \underline{0}) \\ &= P(T_2 \geq c_2 \text{ and } T_1 \geq c_1 | \underline{\alpha}; \underline{\beta} = \underline{0}) \\ &\quad + P(T_2 < c_2 \text{ and } K \geq c_3 | \underline{\alpha}; \underline{\beta} = \underline{0}) \end{aligned}$$

and the power of TK is

$$\begin{aligned} \gamma_N(\underline{\alpha}; \underline{\beta}) &= P(\text{reject } H_{10} | \underline{\alpha}; \underline{\beta} \neq \underline{0}) \\ &= P(T_2 \geq c_2 \text{ and } T_1 \geq c_1 | \underline{\alpha}; \underline{\beta} \neq \underline{0}) \\ &\quad + P(T_2 < c_2 \text{ and } K \geq c_3 | \underline{\alpha}; \underline{\beta} \neq \underline{0}) \end{aligned}$$

It is obvious that both  $\alpha'_N$  and  $\gamma'_N$  depend (directly or indirectly) on the levels  $\alpha''_1$ ,  $\alpha''_2$  and  $\alpha''_3$ , the effects  $\underline{\alpha}$  and  $\underline{\beta}$  and on the sample size  $N$ . It is important to determine the influence that the unknown parameter  $\underline{\alpha}$  has upon  $\alpha'_N$  and  $\gamma'_N$ .

We first consider  $\alpha'_N$ . When  $\underline{\alpha} = \underline{0}$  we find that

$$\begin{aligned}\alpha'_N(\underline{\alpha} = \underline{0}; \underline{\beta} = \underline{0}) &\approx P(T_2 \geq c_2 | \underline{\alpha} = \underline{0}; \underline{\beta} = \underline{0}) \cdot P(T_1 \geq c_1 | \underline{\alpha} = \underline{0}; \underline{\beta} = \underline{0}) \\ &\quad + P(T_2 < c_2 | \underline{\alpha} = \underline{0}; \underline{\beta} = \underline{0}) \cdot P(K \geq c_3 | \underline{\alpha} = \underline{0}; \underline{\beta} = \underline{0}) \\ &\quad \text{because } K \text{ is independent of } T_2 \text{ when } \underline{\alpha} = \underline{0} \\ &\quad \text{(cf. Randles and Hogg (1971)), and the error in the} \\ &\quad \text{first term is very small} \\ &= \alpha'_1 \cdot P(T_2 \geq c_2 | \underline{\alpha} = \underline{0}) + \alpha'_3 [1 - P(T_2 \geq c_2 | \underline{\alpha} = \underline{0})] \\ &\quad \text{because } T_1 \text{ and } K \text{ do not directly involve } H_{20} \\ &\quad \text{and } T_2 \text{ does not directly involve } H_{10} \\ &= \alpha'_3 + (\alpha'_1 - \alpha'_3)\alpha'_2 \\ &\quad \xrightarrow[N \rightarrow \infty]{} \alpha''_1 \quad \text{if } \alpha''_3 = \alpha''_1, \text{ since } \alpha'_i \rightarrow \alpha''_i, \quad i=1,2,3.\end{aligned}$$

When  $\underline{\alpha} \neq \underline{0}$ ,  $T_2$  and  $K$  are dependent and we argue as follows:

For every fixed  $\underline{\alpha} \neq \underline{0}$  and  $N \rightarrow \infty$

$$P(T_2 \geq c_2 | \underline{\alpha}) \rightarrow 1 \quad (T_2 \text{ is consistent}),$$

so that  $P(T_2 \geq c_2 \text{ and } T_1 \geq c_1 | \underline{\alpha}; \underline{\beta} = \underline{0}) \rightarrow P(T_1 \geq \chi^2(k-1, \alpha''_1) | \underline{\beta} = \underline{0}) = \alpha''_1$ , where  $\chi^2(k-1, \alpha''_1)$  denotes the upper  $\alpha''_1$  critical value of the  $\chi^2(k-1)$  distribution,

$$P(T_2 < c_2 \text{ and } K \geq c_3 | \underline{\alpha}; \underline{\beta} = \underline{0}) \rightarrow 0$$

and thus

$$\alpha'_N(\underline{\alpha} \neq \underline{0}; \underline{\beta} = \underline{0}) \rightarrow \alpha''_1.$$

We conclude therefore that  $\alpha_N'(\underline{\alpha}; \underline{\beta} = \underline{0}) \rightarrow \alpha_1''$ , provided  $\alpha_1'' = \alpha_3''$  is chosen.

The problem at hand is to test  $H_{10}: \underline{\beta} = \underline{0}$  at level of significance  $\tau$ . We feel it is reasonable to take  $\alpha_1'' = \alpha_3'' = \tau$  and it is the only case we consider. We now compare  $\alpha_N'$  with  $\tau$  to decide whether TK is robust or not, as the dependence between  $K$  and  $T_2$  when  $\underline{\alpha} \neq \underline{0}$  causes doubt whether TK is distribution-free (cf. Randles and Hogg (1973), pp. 347-348)

- When  $N$  is small enough, the exact distributions of  $T_2, T_1$  and/or  $K$  are available and we then choose the values of  $\alpha_2', \alpha_1'$  and  $\alpha_3'$  which respectively minimise  $|\alpha_2'' - \alpha_2'|$ ,  $|\tau - \alpha_1'|$  and  $|\tau - \alpha_3'|$ . The deviation  $|\alpha_1' - \alpha_3'|$  is mostly so small that  $\alpha_2'$  does not have a marked effect on  $\alpha_N'$ . The pretest procedure will more often than not be applied in situations where the row effects  $\alpha_i$  are not discernible, or not too big, and then  $\min(\alpha_1', \alpha_3') < \alpha_N' < \max(\alpha_1', \alpha_3')$ . Since both the Friedman and Kruskal-Wallis tests are robust, TK will be robust.
- We note that FU is not necessarily robust, since  $F_1$  and  $F_2$  are dependent when  $\underline{\alpha} = \underline{0}$  and  $N$  is small. The size of FU is also greatly influenced by amongst other things the choice of  $\alpha_2''$  (cf. Paul (1950), Bozovich, *et al.* (1956) and Mead, *et al.* (1975)).
- For  $N$  large,  $\alpha_1' \approx \tau \approx \alpha_3'$  so that  $\alpha_N' \approx \tau$  and TK is robust.

Denote the power of  $T_2(T_1, K)$  by  $\gamma_{2N}(\gamma_{1N}, \gamma_{3N})$ . Analogous to  $\alpha_N'$ , the power of TK, when  $\underline{\alpha} = \underline{0}$  and  $\underline{\beta} \neq \underline{0}$ , is

$$\begin{aligned} \gamma_N(\underline{\alpha} = \underline{0}; \underline{\beta} \neq \underline{0}) &\approx P(T_2 \geq c_2 | \underline{\alpha} = \underline{0}) \cdot P(T_1 \geq c_1 | \underline{\beta}) \\ &\quad + P(T_2 < c_2 | \underline{\alpha} = \underline{0}) \cdot P(K \geq c_3 | \underline{\beta}) \\ &= \gamma_{3N} + (\gamma_{1N} - \gamma_{3N})\alpha_2'. \end{aligned}$$

This result also holds approximately for  $\underline{\alpha}$  close to  $\underline{0}$ .

The consistency of the TK test follows directly from that of the Kruskal-Wallis and Friedman tests, so that for every fixed  $\underline{\alpha}$  and  $\underline{\beta} \neq \underline{0}$ ,

$\lim_{N \rightarrow \infty} \gamma_N(\underline{\alpha}; \underline{\beta}) = 1$ . For the asymptotic behaviour of  $\gamma_N$  we therefore consider a sequence  $\{H_N\}$  of contiguous alternative hypotheses, where

$$H_N: F_{ij,N}(x) = F(x - \mu - \alpha_i - \frac{\beta_j}{\sqrt{N}})$$

and  $\underline{\alpha}$  and  $\underline{\beta}$  are vectors of fixed real numbers. Under  $\{H_N\}$  and for  $N \rightarrow \infty$

$$\gamma_N(\underline{\alpha}; \underline{\beta}) = P[T_2 > c_2, T_1 \geq c_1 | \underline{\alpha}; N^{-\frac{1}{2}} \underline{\beta}] + P[T_2 < c_2, K \geq c_3 | \underline{\alpha}; N^{-\frac{1}{2}} \underline{\beta}]$$

$$\xrightarrow{N} \gamma(\underline{\beta})$$

$$= \gamma_1 \cdot I(\underline{\alpha} \neq \underline{0}) + \gamma_3 \cdot I(\underline{\alpha} = \underline{0}), \text{ where } \gamma_i = \lim_{N \rightarrow \infty} \gamma_{iN}, i = 1, 3.$$

When  $\underline{\alpha} = \underline{0}$ , we know that  $K$  is more powerful than  $T_1$  and one would therefore have to recommend a small value for  $\alpha_2''$  to ensure good power for  $TK$ . However, sampling studies show that  $K$  is only slightly more powerful than  $T_1$  and the effect of  $\alpha_2''$  on  $\gamma_N$  cannot be separated from sampling variation (see §5 and the tables).

If  $\underline{\alpha} \neq \underline{0}$ , we find that  $T_1$  is often substantially more powerful than  $K$ . For  $N$  large,  $\gamma_N$  will be close to  $\gamma_1$ .

Clearly  $\gamma_N$  will often lie between  $\gamma_{1N}$  and  $\gamma_{3N}$ , which in some cases differ quite substantially. Furthermore, sampling studies indicate that  $\gamma_N$  will practically always lie close to the largest of  $\gamma_{1N}$  and  $\gamma_{3N}$  (see §5).

#### 4. ASYMPTOTIC RELATIVE EFFICIENCY

We will now use  $\lambda_1^2(\underline{\beta})$  and  $\lambda_1'^2(\underline{\beta})$  to stress the dependence of the non-centrality parameters on  $\underline{\beta}$ .  $G(\varphi, \alpha_1'')$  is the exceedance probability of a  $\chi^2(k-1, \varphi)$  statistic with  $\alpha_1'' (= \alpha_3'')$  the relevant level of significance.

Under  $\{H_N\}$  we found the power of procedure  $TK$  (whether  $\underline{\alpha} = \underline{0}$  or

$\underline{\alpha} \neq \underline{0}$  to be (see §3)

$$\begin{aligned}\gamma_N(\underline{\alpha}; \underline{\beta}) &\xrightarrow[N]{N} \gamma(\underline{\beta}) \\ &= G(\lambda_1^2(\underline{\beta}), \alpha_1'').\end{aligned}$$

The power of the parametric procedure FU is similarly found to be

$$\begin{aligned}\gamma_N^\square(\underline{\alpha}; \underline{\beta}) &= P[F_2 \geq e_2, F_1 \geq e_1 | \underline{\alpha}; N^{-\frac{1}{2}} \underline{\beta}] + P[F_2 < e_2, U \geq e_3 | \underline{\alpha}; N^{-\frac{1}{2}} \underline{\beta}] \\ &\xrightarrow[N]{N} \gamma^\square(\underline{\beta}) \\ &= G(\lambda_1^2(\underline{\beta}), \alpha_1'') \quad (\text{due to the consistency of } F_2).\end{aligned}$$

Consider the sequence  $\{M(N)\}$  of positive integers such that  $\lim_{N \rightarrow \infty} \frac{M(N)}{N} = d^2$  for  $0 < d < \infty$ , and let  $N^{-\frac{1}{2}} \underline{\beta} = \underline{\theta}$ , then

$$\begin{aligned}\gamma_{M(N)}^\square(\underline{\alpha}; \underline{\beta}) &= \gamma_{M(N)}^\square[\underline{\alpha}; \{M(N)\}^{\frac{1}{2}} \underline{\theta}] \\ &\approx \gamma_N^\square[\underline{\alpha}; N^{\frac{1}{2}} d \underline{\theta}] \quad \text{for } N \text{ large enough} \\ &= \gamma_N^\square[\underline{\alpha}; d \underline{\beta}] \\ &\rightarrow G(\lambda_1^2(d \underline{\beta}), \alpha_1'')\end{aligned}$$

which will be equal to  $\lim \gamma_N(\underline{\alpha}, \underline{\beta})$ , provided  $\lambda_1^2(\underline{\beta}) = \lambda_1^2(d \underline{\beta})$  (cf. Saleh and Sen (1982)).

From this equality we have

$$\frac{12}{k} \left( \int f^2(x) dx \right)^2 \Sigma_j \beta_j^2 = \frac{1}{k \sigma^2} \Sigma_j (d \beta_j)^2$$

$$\text{i.e. } d^2 = 12 \sigma^2 \left( \int f^2(x) dx \right)^2.$$

This means that the asymptotic relative efficiency of TK with respect to FU

is  $12 \sigma^2 \left[ \int_{-\infty}^{+\infty} f^2(x) dx \right]^2$ , the well-known a.r.e. of the Kruskal-Wallis test with

respect to the F-test.

Although the foregoing results are asymptotically valid for  $c$  tending to infinity, we know that in practice  $c$  need not necessarily be very large. It is, however, important to know how TK compares with other tests in the case of small values of  $m$ ,  $k$  and  $c$ . Therefore an extensive Monte Carlo study has been performed and the main conclusions are now given.

## 5. SMALL SAMPLE STUDY

An extensive simulation study has been performed in which the values of  $m$ ,  $k$ ,  $c$  and  $\alpha_2''$  have been varied for each of the distributions considered. Every run consisted of 1 000 repetitions. Due to the computer time consumed by every run of the program, it was unfeasible to use more repetitions. Although the results are given to three decimal places, it cannot be considered accurate beyond the second decimal, and even the latter may be in error in a small percentage of cases. The conclusions drawn are, however, based on a large variety of cases and we are convinced that they are valid.

Firstly, the value of  $\alpha_2''$ , the level of significance of the preliminary test, was varied from 0,05 up to 0,50. Although it appeared theoretically that  $\alpha_2''$  could have an effect on the powers of the pretest procedures, this was so small that it could not be separated from sampling variation. (Gans (1981) made the same observation when he considered a preliminary test for the comparison of two sample means.) We thus decided to use  $\alpha_2'' = 0,05$  throughout.

This study showed that the empirical powers of the Friedman and Kruskal-Wallis tests can differ quite substantially (cf. tables 2,5-8). It was found throughout the study that the empirical power of TK is practically always close to the largest of the afore-mentioned powers.

In the case of the uniform distribution the FU test showed a tendency to be liberal sometimes, but as this could not be separated from sampling variation, we are fairly convinced that for all practical purposes FU is robust for all

distributions considered, as is the case with TK. In tables 1-5 the empirical sizes and empirical powers of TK and FU are given for one of the cases considered, namely  $m=k=3$ ,  $c=5$  and  $\alpha_2'' = 0.05$ . The empirical powers of the Friedman, Kruskal-Wallis and corresponding F-tests are also given.

In the case of the normal distribution, TK is generally slightly less powerful than FU, as could be expected, and in the case of the uniform distribution the difference is larger, although not overwhelmingly so.

For the  $\chi^2(4)$  distribution TK is generally more powerful than FU and for the lognormal and exponential distributions TK is appreciably more powerful than FU (cf. tables 2 and 3).

TK was also compared to a test based on aligned ranks, as given by Mehra and Sarangi (1967) (which is a generalisation of a Hodges and Lehmann test). The observations were aligned by using the median, which gave results similar to that of the mean. The test statistic is

$$QM = \left[ \frac{kc-1}{c} \sum_j (R_{.j} - cm(N+1)/2)^2 \right] / \left[ \sum_{ijh} (R_{ijh} - \bar{R}_{i..})^2 \right]$$

where ranks are allocated to the aligned observations jointly. It should be noted that QM is distribution-free only when the configuration of the aligned ranks in the various blocks, is given. However, the conditional null distribution of QM tends to the  $\chi^2(k-1)$  distribution for all configurations.

TK is generally just as powerful as QM when row effects are present, but otherwise tends to be more powerful than QM (cf. tables 6, 7 and 8).

## 6. CONCLUSION

In situations where the researcher is unable to judge whether he should use the Friedman or Kruskal-Wallis test, the pretest procedure has this decision built into it and has been found to reap the benefit of the best test under the prevailing circumstances.

This study has clearly shown that: (a) for this setup the use of a pretest procedure in block designs is worthwhile in terms of power gained; and (b) if any doubt exists regarding the type of distribution involved, the nonparametric pretest procedure is generally better than the parametric one.

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TABLE 1  
EMPIRICAL SIZES AND POWERS OF TESTS FOR THE CASE  $(m, k, c) = (3, 3, 5)$ :  
NORMAL DISTRIBUTION

1.1  $\underline{\alpha} = \underline{0}, \underline{\beta} = \underline{0}$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0,01  | 0,05  | 0,10  |
| $T_1$ | 0,013 | 0,052 | 0,100 |
| KW    | 0,011 | 0,048 | 0,105 |
| TK    | 0,011 | 0,050 | 0,109 |
| $F_1$ | 0,009 | 0,050 | 0,104 |
| U     | 0,010 | 0,051 | 0,101 |
| FU    | 0,011 | 0,052 | 0,104 |

1.2  $\underline{\alpha} = (-0,5 \ 0 \ 0,5), \underline{\beta} = \underline{0}$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0,01  | 0,05  | 0,10  |
| $T_1$ | 0,008 | 0,051 | 0,110 |
| KW    | 0,005 | 0,034 | 0,080 |
| TK    | 0,009 | 0,051 | 0,107 |
| $F_1$ | 0,011 | 0,057 | 0,118 |
| U     | 0,003 | 0,033 | 0,080 |
| FU    | 0,011 | 0,056 | 0,118 |

1.3  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,3 \ 0 \ 0,3)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,097 | 0,232 | 0,350 |
| KW    | 0,088 | 0,244 | 0,367 |
| TK    | 0,094 | 0,246 | 0,368 |
| $F_1$ | 0,110 | 0,264 | 0,386 |
| U     | 0,111 | 0,261 | 0,381 |
| FU    | 0,115 | 0,264 | 0,383 |

1.4  $\underline{\alpha} = (-0,5 \ 0 \ 0,5), \underline{\beta} = (-0,3 \ 0 \ 0,3)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,083 | 0,238 | 0,355 |
| KW    | 0,051 | 0,189 | 0,309 |
| TK    | 0,081 | 0,238 | 0,356 |
| $F_1$ | 0,105 | 0,267 | 0,396 |
| U     | 0,059 | 0,196 | 0,313 |
| FU    | 0,101 | 0,261 | 0,387 |

1.5  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,5 \ 0 \ 0,5)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,363 | 0,610 | 0,715 |
| KW    | 0,336 | 0,617 | 0,743 |
| TK    | 0,342 | 0,619 | 0,746 |
| $F_1$ | 0,399 | 0,648 | 0,768 |
| U     | 0,400 | 0,654 | 0,769 |
| FU    | 0,402 | 0,655 | 0,772 |

1.6  $\underline{\alpha} = (-0,5 \ 0 \ 0,5), \underline{\beta} = (-0,5 \ 0 \ 0,5)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,336 | 0,593 | 0,709 |
| KW    | 0,239 | 0,533 | 0,662 |
| TK    | 0,333 | 0,596 | 0,710 |
| $F_1$ | 0,407 | 0,654 | 0,751 |
| U     | 0,288 | 0,578 | 0,692 |
| FU    | 0,401 | 0,646 | 0,744 |

1.7  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,75 \ 0 \ 0,75)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,788 | 0,937 | 0,969 |
| KW    | 0,790 | 0,945 | 0,974 |
| TK    | 0,794 | 0,947 | 0,975 |
| $F_1$ | 0,828 | 0,967 | 0,980 |
| U     | 0,837 | 0,965 | 0,982 |
| FU    | 0,838 | 0,966 | 0,982 |

1.8  $\underline{\alpha} = (-0,5 \ 0 \ 0,5), \underline{\beta} = (-0,75 \ 0 \ 0,75)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,773 | 0,906 | 0,952 |
| KW    | 0,693 | 0,884 | 0,940 |
| TK    | 0,753 | 0,906 | 0,955 |
| $F_1$ | 0,817 | 0,939 | 0,971 |
| U     | 0,753 | 0,912 | 0,956 |
| FU    | 0,808 | 0,936 | 0,971 |

TABLE 2

EMPIRICAL SIZES AND POWERS OF TESTS FOR THE CASE  $(m, k, c) = (3, 3, 5)$ :  
LOGNORMAL DISTRIBUTION

1.1  $\alpha = 0, \beta = 0$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0,01  | 0,05  | 0,10  |
| $T_1$ | 0,008 | 0,049 | 0,091 |
| KW    | 0,004 | 0,045 | 0,089 |
| TK    | 0,005 | 0,047 | 0,092 |
| $F_1$ | 0,004 | 0,040 | 0,072 |
| U     | 0,005 | 0,039 | 0,070 |
| FU    | 0,005 | 0,039 | 0,071 |

1.3  $\alpha = 0, \beta = (-0,25 \ 0 \ 0,25)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,083 | 0,239 | 0,347 |
| KW    | 0,084 | 0,238 | 0,360 |
| TK    | 0,085 | 0,241 | 0,364 |
| $F_1$ | 0,020 | 0,097 | 0,165 |
| U     | 0,020 | 0,096 | 0,161 |
| FU    | 0,021 | 0,098 | 0,164 |

1.5  $\alpha = 0, \beta = (-0,45 \ 0 \ 0,45)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,303 | 0,560 | 0,681 |
| KW    | 0,302 | 0,583 | 0,706 |
| TK    | 0,307 | 0,578 | 0,707 |
| $F_1$ | 0,085 | 0,220 | 0,336 |
| U     | 0,080 | 0,224 | 0,336 |
| FU    | 0,081 | 0,226 | 0,338 |

1.7  $\alpha = 0, \beta = (-0,7 \ 0 \ 0,7)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,635 | 0,848 | 0,914 |
| KW    | 0,632 | 0,859 | 0,926 |
| TK    | 0,633 | 0,860 | 0,926 |
| $F_1$ | 0,242 | 0,440 | 0,549 |
| U     | 0,243 | 0,442 | 0,552 |
| FU    | 0,244 | 0,443 | 0,554 |

1.2  $\alpha = (-0,45 \ 0 \ 0,45), \beta = 0$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0,01  | 0,05  | 0,10  |
| $T_1$ | 0,013 | 0,056 | 0,112 |
| KW    | 0,007 | 0,037 | 0,085 |
| TK    | 0,012 | 0,056 | 0,111 |
| $F_1$ | 0,004 | 0,040 | 0,106 |
| U     | 0,004 | 0,040 | 0,082 |
| FU    | 0,004 | 0,043 | 0,097 |

1.4  $\alpha = (-0,45 \ 0 \ 0,45),$   
 $\beta = (-0,25 \ 0 \ 0,25)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,077 | 0,237 | 0,367 |
| KW    | 0,036 | 0,147 | 0,258 |
| TK    | 0,067 | 0,215 | 0,340 |
| $F_1$ | 0,018 | 0,093 | 0,173 |
| U     | 0,012 | 0,075 | 0,159 |
| FU    | 0,018 | 0,089 | 0,171 |

1.6  $\alpha = (-0,45 \ 0 \ 0,45),$   
 $\beta = (-0,45 \ 0 \ 0,45)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,318 | 0,579 | 0,703 |
| KW    | 0,176 | 0,457 | 0,593 |
| TK    | 0,289 | 0,554 | 0,674 |
| $F_1$ | 0,070 | 0,197 | 0,310 |
| U     | 0,062 | 0,179 | 0,287 |
| FU    | 0,071 | 0,193 | 0,304 |

1.8  $\alpha = (-0,45 \ 0 \ 0,45),$   
 $\beta = (-0,7 \ 0 \ 0,7)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,669 | 0,855 | 0,921 |
| KW    | 0,548 | 0,815 | 0,891 |
| TK    | 0,639 | 0,856 | 0,916 |
| $F_1$ | 0,259 | 0,464 | 0,576 |
| U     | 0,228 | 0,437 | 0,554 |
| FU    | 0,255 | 0,455 | 0,572 |

TABLE 3  
EMPIRICAL SIZES AND POWERS OF TESTS FOR THE CASE  $(m, k, c) = (3, 3, 5)$ :  
EXPONENTIAL DISTRIBUTION

| 1.1 $\underline{\alpha} = \underline{0}, \underline{\beta} = \underline{0}$      |       |       |       | 1.2 $\underline{\alpha} = (-0,2 \ 0 \ 0,2), \underline{\beta} = \underline{0}$      |       |       |       |
|----------------------------------------------------------------------------------|-------|-------|-------|-------------------------------------------------------------------------------------|-------|-------|-------|
| Level                                                                            |       |       |       | Level                                                                               |       |       |       |
| Test                                                                             | 0,01  | 0,05  | 0,10  | Test                                                                                | 0,01  | 0,05  | 0,10  |
| $T_1$                                                                            | 0,015 | 0,049 | 0,092 | $T_1$                                                                               | 0,010 | 0,047 | 0,084 |
| KW                                                                               | 0,012 | 0,051 | 0,107 | KW                                                                                  | 0,007 | 0,039 | 0,076 |
| TK                                                                               | 0,012 | 0,053 | 0,108 | TK                                                                                  | 0,007 | 0,049 | 0,087 |
| $F_1$                                                                            | 0,009 | 0,063 | 0,108 | $F_1$                                                                               | 0,006 | 0,046 | 0,093 |
| U                                                                                | 0,010 | 0,059 | 0,112 | U                                                                                   | 0,006 | 0,041 | 0,085 |
| FU                                                                               | 0,010 | 0,062 | 0,115 | FU                                                                                  | 0,006 | 0,047 | 0,093 |
| 1.3 $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,08 \ 0 \ 0,8)$  |       |       |       | 1.4 $\underline{\alpha} = (-0,2 \ 0 \ 0,2), \underline{\beta} = (-0,08 \ 0 \ 0,8)$  |       |       |       |
| $T_1$                                                                            | 0,015 | 0,073 | 0,125 | $T_1$                                                                               | 0,018 | 0,074 | 0,129 |
| KW                                                                               | 0,015 | 0,073 | 0,134 | KW                                                                                  | 0,015 | 0,065 | 0,115 |
| TK                                                                               | 0,015 | 0,077 | 0,137 | TK                                                                                  | 0,019 | 0,070 | 0,132 |
| $F_1$                                                                            | 0,010 | 0,053 | 0,107 | $F_1$                                                                               | 0,014 | 0,061 | 0,107 |
| U                                                                                | 0,010 | 0,055 | 0,108 | U                                                                                   | 0,012 | 0,052 | 0,104 |
| FU                                                                               | 0,010 | 0,055 | 0,109 | FU                                                                                  | 0,014 | 0,058 | 0,107 |
| 1.5 $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,2 \ 0 \ 0,2)$   |       |       |       | 1.6 $\underline{\alpha} = (-0,2 \ 0 \ 0,2), \underline{\beta} = (-0,2 \ 0 \ 0,2)$   |       |       |       |
| $T_1$                                                                            | 0,097 | 0,278 | 0,376 | $T_1$                                                                               | 0,099 | 0,270 | 0,398 |
| KW                                                                               | 0,088 | 0,277 | 0,399 | KW                                                                                  | 0,067 | 0,231 | 0,359 |
| TK                                                                               | 0,091 | 0,280 | 0,400 | TK                                                                                  | 0,086 | 0,270 | 0,385 |
| $F_1$                                                                            | 0,046 | 0,147 | 0,244 | $F_1$                                                                               | 0,039 | 0,155 | 0,264 |
| U                                                                                | 0,046 | 0,148 | 0,244 | U                                                                                   | 0,033 | 0,149 | 0,250 |
| FU                                                                               | 0,046 | 0,150 | 0,245 | FU                                                                                  | 0,038 | 0,157 | 0,261 |
| 1.7 $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,35 \ 0 \ 0,35)$ |       |       |       | 1.8 $\underline{\alpha} = (-0,2 \ 0 \ 0,2), \underline{\beta} = (-0,35 \ 0 \ 0,35)$ |       |       |       |
| $T_1$                                                                            | 0,324 | 0,588 | 0,717 | $T_1$                                                                               | 0,346 | 0,597 | 0,722 |
| KW                                                                               | 0,316 | 0,613 | 0,752 | KW                                                                                  | 0,295 | 0,579 | 0,708 |
| TK                                                                               | 0,319 | 0,616 | 0,756 | TK                                                                                  | 0,316 | 0,596 | 0,722 |
| $F_1$                                                                            | 0,171 | 0,402 | 0,513 | $F_1$                                                                               | 0,190 | 0,400 | 0,555 |
| U                                                                                | 0,169 | 0,391 | 0,514 | U                                                                                   | 0,170 | 0,380 | 0,538 |
| FU                                                                               | 0,172 | 0,398 | 0,517 | FU                                                                                  | 0,179 | 0,392 | 0,552 |

TABLE 4

EMPIRICAL SIZES AND POWERS OF TESTS FOR THE CASE  $(m, k, c) = (3, 3, 5)$ : $\chi^2(4)$  DISTRIBUTION

| 1.1 $\underline{\alpha} = \underline{0}, \underline{\beta} = \underline{0}$    |       |       |       | 1.2 $\underline{\alpha} = (-0,9 \ 0 \ 0,9), \underline{\beta} = \underline{0}$    |       |       |       |
|--------------------------------------------------------------------------------|-------|-------|-------|-----------------------------------------------------------------------------------|-------|-------|-------|
| <u>Level</u>                                                                   |       |       |       | <u>Level</u>                                                                      |       |       |       |
| Test                                                                           | 0,01  | 0,05  | 0,10  | Test                                                                              | 0,01  | 0,05  | 0,10  |
| $T_1$                                                                          | 0,008 | 0,048 | 0,094 | $T_1$                                                                             | 0,013 | 0,048 | 0,109 |
| KW                                                                             | 0,006 | 0,051 | 0,094 | KW                                                                                | 0,008 | 0,040 | 0,083 |
| TK                                                                             | 0,006 | 0,053 | 0,097 | TK                                                                                | 0,013 | 0,049 | 0,102 |
| $F_1$                                                                          | 0,008 | 0,047 | 0,097 | $F_1$                                                                             | 0,012 | 0,052 | 0,106 |
| U                                                                              | 0,010 | 0,050 | 0,098 | U                                                                                 | 0,007 | 0,046 | 0,091 |
| FU                                                                             | 0,010 | 0,050 | 0,100 | FU                                                                                | 0,010 | 0,050 | 0,106 |
| 1.3 $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,6 \ 0 \ 0,6)$ |       |       |       | 1.4 $\underline{\alpha} = (-0,9 \ 0 \ 0,9), \underline{\beta} = (-0,6 \ 0 \ 0,6)$ |       |       |       |
| $T_1$                                                                          | 0,067 | 0,190 | 0,304 | $T_1$                                                                             | 0,060 | 0,193 | 0,321 |
| KW                                                                             | 0,069 | 0,197 | 0,319 | KW                                                                                | 0,039 | 0,146 | 0,269 |
| TK                                                                             | 0,069 | 0,199 | 0,324 | TK                                                                                | 0,050 | 0,181 | 0,310 |
| $F_1$                                                                          | 0,061 | 0,173 | 0,291 | $F_1$                                                                             | 0,052 | 0,169 | 0,276 |
| U                                                                              | 0,058 | 0,174 | 0,283 | U                                                                                 | 0,035 | 0,145 | 0,238 |
| FU                                                                             | 0,058 | 0,177 | 0,285 | FU                                                                                | 0,048 | 0,167 | 0,267 |
| 1.5 $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,9 \ 0 \ 0,9)$ |       |       |       | 1.6 $\underline{\alpha} = (-0,9 \ 0 \ 0,9), \underline{\beta} = (-0,9 \ 0 \ 0,9)$ |       |       |       |
| $T_1$                                                                          | 0,162 | 0,377 | 0,526 | $T_1$                                                                             | 0,180 | 0,404 | 0,536 |
| KW                                                                             | 0,165 | 0,402 | 0,526 | KW                                                                                | 0,132 | 0,356 | 0,491 |
| TK                                                                             | 0,168 | 0,406 | 0,530 | TK                                                                                | 0,170 | 0,400 | 0,528 |
| $F_1$                                                                          | 0,130 | 0,323 | 0,448 | $F_1$                                                                             | 0,155 | 0,339 | 0,468 |
| U                                                                              | 0,129 | 0,328 | 0,451 | U                                                                                 | 0,128 | 0,313 | 0,437 |
| FU                                                                             | 0,130 | 0,329 | 0,454 | FU                                                                                | 0,153 | 0,337 | 0,463 |
| 1.7 $\underline{\alpha} = \underline{0}, \underline{\beta} = (-1,6 \ 0 \ 1,6)$ |       |       |       | 1.8 $\underline{\alpha} = (-0,9 \ 0 \ 0,9), \underline{\beta} = (-1,6 \ 0 \ 1,6)$ |       |       |       |
| $T_1$                                                                          | 0,654 | 0,860 | 0,922 | $T_1$                                                                             | 0,686 | 0,872 | 0,926 |
| KW                                                                             | 0,655 | 0,878 | 0,939 | KW                                                                                | 0,625 | 0,852 | 0,928 |
| TK                                                                             | 0,659 | 0,882 | 0,938 | TK                                                                                | 0,676 | 0,873 | 0,933 |
| $F_1$                                                                          | 0,577 | 0,794 | 0,891 | $F_1$                                                                             | 0,598 | 0,807 | 0,878 |
| U                                                                              | 0,585 | 0,797 | 0,888 | U                                                                                 | 0,556 | 0,788 | 0,867 |
| FU                                                                             | 0,590 | 0,800 | 0,890 | FU                                                                                | 0,586 | 0,808 | 0,878 |

TABLE 5

EMPIRICAL SIZES AND POWERS OF TESTS FOR THE CASE  $(m, k, c) = (3, 3, 5)$ :  
UNIFORM DISTRIBUTION

1.1  $\underline{\alpha} = 0, \underline{\beta} = 0$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0, 01 | 0, 05 | 0, 10 |
| $T_1$ | 0,015 | 0,049 | 0,092 |
| KW    | 0,012 | 0,051 | 0,107 |
| TK    | 0,012 | 0,053 | 0,108 |
| $F_1$ | 0,018 | 0,054 | 0,112 |
| U     | 0,017 | 0,053 | 0,112 |
| FU    | 0,017 | 0,054 | 0,113 |

1.2  $\underline{\alpha} = (-0,17 \ 0 \ 0,17), \underline{\beta} = 0$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0, 01 | 0, 05 | 0, 10 |
| $T_1$ | 0,010 | 0,047 | 0,084 |
| KW    | 0,003 | 0,027 | 0,054 |
| TK    | 0,010 | 0,049 | 0,082 |
| $F_1$ | 0,012 | 0,046 | 0,088 |
| U     | 0,004 | 0,026 | 0,057 |
| FU    | 0,011 | 0,046 | 0,088 |

1.3  $\underline{\alpha} = 0, \underline{\beta} = (-0,12 \ 0 \ 0,12)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,198 | 0,438 | 0,579 |
| KW    | 0,189 | 0,451 | 0,599 |
| TK    | 0,195 | 0,457 | 0,604 |
| $F_1$ | 0,257 | 0,518 | 0,645 |
| U     | 0,252 | 0,521 | 0,645 |
| FU    | 0,259 | 0,526 | 0,649 |

1.4  $\underline{\alpha} = (-0,17 \ 0 \ 0,17),$   
 $\underline{\beta} = (-0,12 \ 0 \ 0,12)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,202 | 0,410 | 0,544 |
| KW    | 0,108 | 0,307 | 0,453 |
| TK    | 0,190 | 0,404 | 0,542 |
| $F_1$ | 0,227 | 0,478 | 0,606 |
| U     | 0,144 | 0,346 | 0,516 |
| FU    | 0,223 | 0,473 | 0,604 |

1.5  $\underline{\alpha} = 0, \underline{\beta} = (-0,17 \ 0 \ 0,17)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,468 | 0,720 | 0,827 |
| KW    | 0,463 | 0,747 | 0,840 |
| TK    | 0,468 | 0,748 | 0,842 |
| $F_1$ | 0,558 | 0,814 | 0,889 |
| U     | 0,563 | 0,809 | 0,890 |
| FU    | 0,565 | 0,813 | 0,891 |

1.6  $\underline{\alpha} = (-0,17 \ 0 \ 0,17),$   
 $\underline{\beta} = (-0,17 \ 0 \ 0,17)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,446 | 0,710 | 0,810 |
| KW    | 0,313 | 0,623 | 0,755 |
| TK    | 0,431 | 0,711 | 0,805 |
| $F_1$ | 0,558 | 0,805 | 0,888 |
| U     | 0,391 | 0,712 | 0,821 |
| FU    | 0,548 | 0,800 | 0,885 |

1.7  $\underline{\alpha} = 0, \underline{\beta} = (-0,2 \ 0 \ 0,2)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,661 | 0,853 | 0,920 |
| KW    | 0,637 | 0,875 | 0,937 |
| TK    | 0,645 | 0,877 | 0,938 |
| $F_1$ | 0,758 | 0,927 | 0,964 |
| U     | 0,760 | 0,926 | 0,966 |
| FU    | 0,762 | 0,928 | 0,966 |

1.8  $\underline{\alpha} = (-0,17 \ 0 \ 0,17),$   
 $\underline{\beta} = (-0,2 \ 0 \ 0,2)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,659 | 0,858 | 0,920 |
| KW    | 0,509 | 0,799 | 0,894 |
| TK    | 0,648 | 0,860 | 0,930 |
| $F_1$ | 0,768 | 0,928 | 0,966 |
| U     | 0,631 | 0,873 | 0,936 |
| FU    | 0,764 | 0,926 | 0,964 |

TABLE 6

EMPIRICAL SIZES AND POWERS OF NONPARAMETRIC TESTS  
FOR THE CASE  $(m, k, c) = (5, 3, 2)$ : NORMAL DISTRIBUTION

1.1  $\underline{\alpha} = 0, \underline{\beta} = 0$

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0,01  | 0,05  | 0,10  |
| $T_1$ | 0,004 | 0,045 | 0,101 |
| KW    | 0,006 | 0,037 | 0,101 |
| TK    | 0,005 | 0,040 | 0,111 |
| QM    | 0,005 | 0,048 | 0,090 |

1.2  $\underline{\alpha} = (-0,6 -0,2 0 0,2 0,6), \underline{\beta} = 0$

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0,01  | 0,05  | 0,10  |
| $T_1$ | 0,008 | 0,035 | 0,073 |
| KW    | 0,002 | 0,026 | 0,061 |
| TK    | 0,006 | 0,032 | 0,073 |
| QM    | 0,008 | 0,041 | 0,089 |

1.3  $\underline{\alpha} = 0, \underline{\beta} = (-0,3 0 0,3)$

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,049 | 0,173 | 0,250 |
| KW    | 0,050 | 0,186 | 0,293 |
| TK    | 0,053 | 0,190 | 0,296 |
| QM    | 0,034 | 0,161 | 0,256 |

1.4  $\underline{\alpha} = (-0,6 -0,2 0 0,2 0,6),$   
 $\underline{\beta} = (-0,3 0 0,3)$

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,056 | 0,172 | 0,270 |
| KW    | 0,029 | 0,140 | 0,236 |
| TK    | 0,049 | 0,172 | 0,264 |
| QM    | 0,047 | 0,177 | 0,272 |

1.5  $\underline{\alpha} = 0, \underline{\beta} = (-0,6 0 0,6)$

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,264 | 0,514 | 0,640 |
| KW    | 0,282 | 0,583 | 0,704 |
| TK    | 0,284 | 0,588 | 0,706 |
| QM    | 0,223 | 0,532 | 0,675 |

1.6  $\underline{\alpha} = (-0,6 -0,2 0 0,2 0,6),$   
 $\underline{\beta} = (-0,6 0 0,6)$

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,276 | 0,545 | 0,673 |
| KW    | 0,223 | 0,522 | 0,683 |
| TK    | 0,267 | 0,565 | 0,722 |
| QM    | 0,246 | 0,555 | 0,684 |

1.7  $\underline{\alpha} = 0, \underline{\beta} = (-0,9 0 0,9)$

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,632 | 0,853 | 0,919 |
| KW    | 0,701 | 0,912 | 0,959 |
| TK    | 0,702 | 0,914 | 0,961 |
| QM    | 0,672 | 0,895 | 0,952 |

1.8  $\underline{\alpha} = (-0,6 -0,2 0 0,2 0,6),$   
 $\underline{\beta} = (-0,9 0 0,9)$

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,659 | 0,859 | 0,918 |
| KW    | 0,611 | 0,880 | 0,941 |
| TK    | 0,667 | 0,888 | 0,944 |
| QM    | 0,686 | 0,904 | 0,954 |

TABLE 7

EMPIRICAL SIZES AND POWERS OF NONPARAMETRIC TESTS  
FOR THE CASE  $(m, k, c) = (5, 3, 2)$ : LOGNORMAL DISTRIBUTION

1.1  $\underline{\alpha} = \underline{0}, \underline{\beta} = \underline{0}$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0, 01 | 0, 05 | 0, 10 |
| $T_1$ | 0,004 | 0,038 | 0,084 |
| KW    | 0,002 | 0,033 | 0,089 |
| TK    | 0,003 | 0,036 | 0,092 |
| QM    | 0,008 | 0,050 | 0,100 |

1.2  $\underline{\alpha} = (-0,5 \ -0,05 \ 0 \ 0,05 \ 0,5),$   
 $\underline{\beta} = \underline{0}$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0, 01 | 0, 05 | 0, 10 |
| $T_1$ | 0,011 | 0,053 | 0,104 |
| KW    | 0,004 | 0,032 | 0,077 |
| TK    | 0,008 | 0,043 | 0,095 |
| QM    | 0,006 | 0,065 | 0,112 |

1.3  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,25 \ 0 \ 0,25)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,035 | 0,147 | 0,230 |
| KW    | 0,036 | 0,163 | 0,259 |
| TK    | 0,038 | 0,167 | 0,263 |
| QM    | 0,029 | 0,117 | 0,220 |

1.4  $\underline{\alpha} = (-0,5 \ -0,05 \ 0 \ 0,05 \ 0,5),$   
 $\underline{\beta} = (-0,25 \ 0 \ 0,25)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,042 | 0,140 | 0,223 |
| KW    | 0,023 | 0,106 | 0,199 |
| TK    | 0,037 | 0,135 | 0,226 |
| QM    | 0,026 | 0,127 | 0,214 |

1.5  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,45 \ 0 \ 0,45)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,141 | 0,361 | 0,473 |
| KW    | 0,155 | 0,396 | 0,543 |
| TK    | 0,159 | 0,408 | 0,550 |
| QM    | 0,115 | 0,332 | 0,464 |

1.6  $\underline{\alpha} = (-0,5 \ -0,05 \ 0 \ 0,05 \ 0,5)$   
 $\underline{\beta} = (-0,45 \ 0 \ 0,45)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,160 | 0,315 | 0,444 |
| KW    | 0,121 | 0,294 | 0,416 |
| TK    | 0,137 | 0,311 | 0,448 |
| QM    | 0,122 | 0,322 | 0,452 |

1.7  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,7 \ 0 \ 0,7)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,365 | 0,623 | 0,740 |
| KW    | 0,397 | 0,688 | 0,803 |
| TK    | 0,401 | 0,687 | 0,801 |
| QM    | 0,303 | 0,579 | 0,705 |

1.8  $\underline{\alpha} = (-0,5 \ -0,05 \ 0 \ 0,05 \ 0,5),$   
 $\underline{\beta} = (-0,7 \ 0 \ 0,7)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,367 | 0,640 | 0,753 |
| KW    | 0,333 | 0,617 | 0,744 |
| TK    | 0,361 | 0,643 | 0,758 |
| QM    | 0,325 | 0,616 | 0,757 |

TABLE 8

EMPIRICAL SIZES AND POWERS OF NONPARAMETRIC TESTS  
FOR THE CASE  $(m, k, c) = (5, 3, 2)$ :  $\chi^2(4)$  DISTRIBUTION

1.1  $\underline{\alpha} = \underline{0}, \underline{\beta} = \underline{0}$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0, 01 | 0, 05 | 0, 10 |
| $T_1$ | 0,008 | 0,038 | 0,073 |
| KW    | 0,006 | 0,044 | 0,088 |
| TK    | 0,007 | 0,046 | 0,094 |
| QM    | 0,006 | 0,043 | 0,097 |

1.2  $\underline{\alpha} = (-0,75 \ -0,1 \ 0 \ 0,1 \ 0,75),$   
 $\underline{\beta} = \underline{0}$ 

| Test  | Level |       |       |
|-------|-------|-------|-------|
|       | 0, 01 | 0, 05 | 0, 10 |
| $T_1$ | 0,006 | 0,052 | 0,098 |
| KW    | 0,006 | 0,045 | 0,091 |
| TK    | 0,009 | 0,049 | 0,094 |
| QM    | 0,007 | 0,052 | 0,084 |

1.3  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-0,75 \ 0 \ 0,75)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,061 | 0,177 | 0,278 |
| KW    | 0,060 | 0,189 | 0,313 |
| TK    | 0,062 | 0,194 | 0,315 |
| QM    | 0,051 | 0,180 | 0,298 |

1.4  $\underline{\alpha} = (-0,75 \ -0,1 \ 0 \ 0,1 \ 0,75),$   
 $\underline{\beta} = (-0,75 \ 0 \ 0,75)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,058 | 0,178 | 0,274 |
| KW    | 0,049 | 0,185 | 0,293 |
| TK    | 0,060 | 0,196 | 0,306 |
| QM    | 0,048 | 0,192 | 0,285 |

1.5  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-1,5 \ 0 \ 1,5)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,292 | 0,535 | 0,665 |
| KW    | 0,311 | 0,624 | 0,738 |
| TK    | 0,314 | 0,626 | 0,738 |
| QM    | 0,267 | 0,574 | 0,709 |

1.6  $\underline{\alpha} = (-0,75 \ -0,1 \ 0 \ 0,1 \ 0,75),$   
 $\underline{\beta} = (-1,5 \ 0 \ 1,5)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,297 | 0,572 | 0,687 |
| KW    | 0,302 | 0,595 | 0,725 |
| TK    | 0,311 | 0,610 | 0,734 |
| QM    | 0,286 | 0,596 | 0,732 |

1.7  $\underline{\alpha} = \underline{0}, \underline{\beta} = (-2,1 \ 0 \ 2,1)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,566 | 0,807 | 0,886 |
| KW    | 0,623 | 0,865 | 0,940 |
| TK    | 0,632 | 0,866 | 0,942 |
| QM    | 0,558 | 0,831 | 0,912 |

1.8  $\underline{\alpha} = (-0,75 \ -0,1 \ 0 \ 0,1 \ 0,75),$   
 $\underline{\beta} = (-2,1 \ 0 \ 2,1)$ 

|       |       |       |       |
|-------|-------|-------|-------|
| $T_1$ | 0,608 | 0,835 | 0,908 |
| KW    | 0,617 | 0,870 | 0,939 |
| TK    | 0,632 | 0,874 | 0,940 |
| QM    | 0,592 | 0,857 | 0,937 |